

Project Title :

Ford GoBike Trip Analysis



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Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn
Date

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1. Project Overview

Introduction

The Ford GoBike Sharing System is a public bicycle-sharing service that provides an affordable, convenient, and sustainable mode of transportation for commuters and recreational riders across the San Francisco Bay Area. Every trip made through the system generates valuable data, including trip duration, station locations, ride timings, and rider demographics. Analyzing this data helps uncover travel patterns, understand customer behavior, and identify opportunities to improve operational efficiency and customer experience.

This project performs a comprehensive Exploratory Data Analysis (EDA) on the 2018 Ford GoBike trip dataset using Python. The analysis involves data cleaning, feature engineering, statistical exploration, and data visualization to transform raw trip records into meaningful business insights. By examining customer demographics, ride trends, station usage, trip durations, and temporal patterns, the project aims to support data-driven decision-making for bike-sharing operations.

Business Problem

Bike-sharing services generate millions of trip records every year, making it difficult to identify meaningful patterns without proper analysis. Understanding when customers ride, where they travel, how long trips last, and which stations experience the highest demand is essential for optimizing fleet management, improving station planning, and enhancing customer satisfaction.

Without data-driven insights, operators may face challenges such as bike shortages at high-demand stations, inefficient redistribution of bicycles, increased maintenance costs, and missed opportunities to improve customer engagement. This project addresses these challenges by analyzing historical trip data to identify key trends and provide actionable business recommendations.

Project Scope

This project covers the complete Exploratory Data Analysis (EDA) workflow using the Ford GoBike 2018 trip dataset. The analysis includes merging monthly datasets, assessing data quality, performing data cleaning, engineering new features, and conducting detailed exploratory analysis through 18 visualizations. The project investigates customer demographics, ride frequency, temporal trends, station performance, trip characteristics, and fleet utilization to generate meaningful insights for business decision-making.

2. Dataset Description

Dataset Overview

The dataset used in this project is the Ford GoBike Trip Data (2018), which contains detailed records of bike-sharing trips made throughout the San Francisco Bay Area. The dataset captures trip-level information such as trip duration, start and end stations, ride timestamps, user type, gender, birth year, and bike identification. Each record represents a single completed bike trip, providing valuable insights into customer behavior and system usage.

To perform a comprehensive annual analysis, twelve monthly datasets were combined into a single dataset. This allowed for the identification of long-term trends, seasonal patterns, and customer usage behaviors across the entire year.

Dataset Information

Attribute	Details
Dataset Name	Ford GoBike Trip Data (2018)
Data Source	Ford GoBike Open Data
File Format	CSV
Time Period	January 2018 – December 2018
Number of Files	12 Monthly CSV Files
Total Records	Approximately 1.86 Million Trips (Before Cleaning)
Primary Tool	Python (Pandas)

Note: The final number of records used for analysis was obtained after removing missing values, duplicate records, and invalid age entries during the data cleaning process.

Dataset Features

The dataset consists of variables that describe each bike trip, including ride information, station details, customer demographics, and membership status.

Feature	Description
duration_sec	Total trip duration in seconds
start_time	Trip start date and time
end_time	Trip end date and time
start_station_id	Unique identifier of the starting station
start_station_name	Name of the starting station
start_station_latitude	Latitude of the starting station
start_station_longitude	Longitude of the starting station
end_station_id	Unique identifier of the destination station
end_station_name	Name of the destination station
end_station_latitude	Latitude of the destination station
end_station_longitude	Longitude of the destination station
bike_id	Unique identifier for each bicycle
user_type	Subscriber or Customer
member_birth_year	Birth year of the rider
member_gender	Gender of the rider

Dataset Characteristics

The dataset contains a mix of numerical, categorical, and datetime variables, making it suitable for performing demographic analysis, time-series analysis, station utilization analysis, and customer behavior analysis. Since the original data was collected monthly, merging all files into a single dataset provided a complete view of bike-sharing activities over the course of one year.

Before analysis, the dataset underwent data quality assessment and preprocessing to ensure consistency, accuracy, and reliability of the results.

3. Project Objectives

Project Objectives

The primary objective of this project is to analyze the **Ford GoBike 2018 trip dataset** to uncover meaningful patterns in customer behavior, ride usage, and operational performance. By applying data cleaning, feature engineering, and exploratory data analysis (EDA), the project transforms raw trip data into actionable business insights that can support data-driven decision-making.

The specific objectives of this project are:

- Analyze customer demographics, including user type, gender, and age distribution.
- Identify ride usage patterns across different months, weekdays, hours, and times of the day.
- Examine trip duration trends and compare ride behaviors among different customer groups.
- Identify the most frequently used bike stations and popular travel routes.
- Evaluate bike utilization to understand fleet usage and operational efficiency.
- Explore relationships between numerical variables using correlation analysis.
- Generate business insights and recommendations to improve fleet management, resource allocation, and customer experience.

Key Business Questions

This analysis aims to answer the following business questions:

1. Who are the primary users of the Ford GoBike system?
2. During which months and time periods is bike-sharing demand the highest?
3. What are the most common trip durations?
4. Which stations experience the highest number of departures and arrivals?
5. What are the most frequently traveled routes?
6. How effectively is the bike fleet being utilized?
7. What actionable insights can help improve operational efficiency and customer satisfaction?

Expected Business Value

The insights generated from this analysis can help bike-sharing operators optimize fleet distribution, improve station planning, reduce operational inefficiencies, and enhance the overall customer experience. By understanding customer behavior and demand patterns, the organization can make informed decisions regarding resource allocation, maintenance scheduling, and service improvements.

4. Data Understanding

Overview

Understanding the dataset is an essential first step in any data analysis project. Before performing data cleaning and exploratory analysis, it is important to examine the dataset's structure, data types, and overall quality. This process helps identify inconsistencies, missing values, duplicate records, and potential issues that may affect the accuracy of the analysis.

The Ford GoBike dataset contains detailed information about each bike-sharing trip, including trip duration, station details, ride timestamps, customer demographics, and membership type. These variables provide valuable information for understanding customer behavior and operational performance.

Dataset Structure

After merging all twelve monthly datasets into a single DataFrame, the dataset was examined to understand its overall structure.

The following checks were performed:

- Examined the total number of rows and columns.
- Reviewed the list of available features.
- Identified the data type of each variable.
- Displayed sample records to verify data consistency.
- Generated descriptive statistics for numerical and categorical variables.

These initial checks provided a clear understanding of the dataset before moving to the data quality assessment stage.

Types of Variables

The dataset contains different types of variables that support various forms of analysis.

Variable Type	Examples
Numerical	Trip Duration, Bike ID, Station ID, Birth Year
Categorical	User Type, Gender, Station Names
Datetime	Start Time, End Time
Geographical	Latitude and Longitude of Start and End Stations

The presence of numerical, categorical, datetime, and geographical variables makes the dataset suitable for demographic analysis, temporal trend analysis, and station utilization analysis.

Initial Observations

The preliminary exploration of the dataset revealed several important observations:

- The dataset contains records for bike trips made throughout the year 2018.
- Most variables were complete and suitable for analysis.
- A few columns contained missing values that required preprocessing.
- Datetime columns needed to be converted into the appropriate format for time-based analysis.
- Customer demographic information enabled segmentation based on user type, gender, and age.
- Station and route information allowed analysis of high-demand locations and travel patterns.

These observations highlighted the need for a structured data quality assessment before proceeding with detailed analysis.

5. Data Cleaning & Feature Engineering

Overview

Before performing exploratory data analysis, the dataset underwent a comprehensive data cleaning and feature engineering process to improve its quality and prepare it for meaningful analysis. Raw datasets often contain missing values, duplicate records, incorrect data types, and inconsistent information that can negatively impact analytical results. Therefore, several preprocessing techniques were applied to ensure the dataset was accurate, consistent, and suitable for business analysis.

Data Cleaning: The following data cleaning steps were performed to improve the overall quality of the dataset:

1. Handling Missing Values

Missing values were identified in station-related columns and customer demographic fields. Records with missing station information were removed, as they were essential for station and route analysis. Missing demographic information, such as birth year and gender, was handled appropriately based on the requirements of the analysis.

2. Removing Duplicate Records

Duplicate records were identified and removed to ensure that each bike trip was represented only once. This prevented duplicate entries from affecting statistical summaries and visualizations.

3. Data Type Conversion

The **Start Time** and **End Time** columns were converted from text format to datetime format, enabling time-based analysis and the creation of additional temporal features.

4. Handling Invalid Age Values

Rider age was calculated using the birth year. Records containing unrealistic age values (less than 18 years or greater than 80 years) were removed to improve the reliability of demographic analysis.

Feature Engineering

To enhance the analytical capabilities of the dataset, several new features were created from the existing variables.

Feature	Description
Trip Duration (Minutes)	Converted trip duration from seconds to minutes for easier interpretation.
Ride Year	Extracted the year from the trip start time.
Ride Month	Extracted the numerical month for monthly trend analysis.
Month Name	Generated month names for better visualization and readability.
Day Name	Extracted the day of the week from the trip date.
Ride Hour	Extracted the hour of the day to analyze peak riding times.
Day Type	Classified rides as Weekday or Weekend .
Age	Calculated rider age using the birth year.

Time of Day	Categorized rides into Morning, Afternoon, Evening, and Night.
Route	Combined start and end station names to identify the most frequently traveled routes.

Benefits of Feature Engineering

Creating new variables significantly improved the depth of analysis by enabling:

- Monthly and seasonal trend analysis.
- Hourly demand pattern analysis.
- Customer segmentation by age and ride timing.
- Weekday versus weekend comparisons.
- Identification of popular travel routes.
- Improved readability and interpretation of visualizations.

7. Exploratory Data Analysis (EDA)

Overview

Exploratory Data Analysis (EDA) was conducted to uncover meaningful patterns, trends, and relationships within the Ford GoBike trip dataset. After completing data cleaning and feature engineering, a total of **18 visualizations** were created using **Matplotlib** and **Seaborn** to analyze customer behavior, trip characteristics, temporal trends, station performance, and fleet utilization.

The visualizations are organized into six major analytical categories to provide a structured understanding of the bike-sharing system and support data-driven business decision-making.

EDA Categories

The exploratory analysis is divided into the following sections:

7.1 Customer Analysis

This section examines the demographic characteristics of riders, including user type, gender distribution, age distribution, and age group segmentation. The objective is to understand the primary customer base and identify the dominant rider segments.

7.2 Time-Based Analysis

This section analyzes ride demand across different time dimensions, including monthly trends, weekly patterns, hourly usage, weekday versus weekend comparisons, and time-of-day analysis. These insights help identify peak demand periods and operational patterns.

7.3 Trip Analysis

This section focuses on ride duration and trip behavior. It examines the distribution of trip durations and compares ride lengths across different customer groups, genders, and age categories.

7.4 Station Analysis

This section evaluates station performance by identifying the busiest start stations, end stations, and the most frequently traveled routes. The findings highlight high-demand locations within the bike-sharing network.

7.5 Bike Utilization Analysis

This section investigates fleet utilization by identifying the most frequently used bicycles. The analysis provides insights into bike usage patterns and supports maintenance and fleet management decisions.

7.6 Correlation Analysis

The final analytical section explores relationships between numerical variables using a correlation heatmap. This helps identify any significant associations that may influence rider behavior or operational performance.

Summary

The Exploratory Data Analysis provides a comprehensive understanding of customer behavior, trip characteristics, demand patterns, station utilization, and fleet performance. The insights obtained from these analyses form the basis for the business recommendations presented later in the report.

8. Business Recommendations

Overview

Based on the insights obtained from the exploratory data analysis, several recommendations can be made to improve the operational efficiency, customer experience, and overall performance of the Ford GoBike bike-sharing system. These recommendations are driven by observed customer behavior, trip patterns, station demand, and fleet utilization.

1. Optimize Bike Distribution During Peak Hours

The analysis identified clear demand peaks during weekday morning and evening commuting hours. Increasing bike availability at high-demand stations before these peak periods can reduce shortages and improve customer satisfaction.

2. Expand Capacity at High-Demand Stations

Stations such as San Francisco Caltrain Station 2 and other major transit hubs consistently recorded the highest number of departures and arrivals. Expanding docking capacity and ensuring adequate bike availability at these locations can improve service efficiency.

3. Improve Weekend Ridership

The majority of trips were recorded on weekdays, indicating that the service is primarily used for commuting. Promotional campaigns, discounted passes, and recreational ride packages can encourage greater weekend usage and improve overall system utilization.

4. Strengthen Integration with Public Transportation

Many of the busiest stations are located near major transit hubs. Improving integration between the bike-sharing system and public transportation can encourage multimodal travel and provide a more convenient commuting experience.

5. Implement Predictive Fleet Maintenance

Bike utilization analysis revealed that certain bicycles are used significantly more frequently than others. Monitoring bike usage and scheduling preventive maintenance based on ride frequency can reduce unexpected breakdowns and extend fleet lifespan.

6. Increase Customer Retention

Subscribers account for the majority of trips, demonstrating the success of the membership model. Introducing loyalty rewards, referral programs, and personalized offers can help retain existing subscribers while encouraging casual customers to convert into long-term members.

7. Use Data-Driven Demand Forecasting

Historical ride patterns can be used to predict future demand across different seasons, weekdays, and hours. Predictive demand forecasting can support more efficient bike redistribution, staffing decisions, and operational planning.

8. Promote Sustainable Urban Mobility

The high volume of short-distance trips demonstrates the importance of bike sharing as an alternative mode of transportation. Continued investment in cycling infrastructure and awareness campaigns can encourage sustainable travel while reducing traffic congestion and environmental impact.

Summary

Implementing these recommendations can help improve fleet management, increase customer satisfaction, optimize operational efficiency, and support the continued growth of the Ford GoBike bike-sharing system through data-driven decision-making.

9. Project Conclusion

This project presented a comprehensive exploratory data analysis of the Ford GoBike 2018 trip dataset using Python. The analysis followed a structured workflow, beginning with data collection, quality assessment, data cleaning, and feature engineering before progressing to exploratory analysis and business interpretation.

A total of 18 visualizations were developed to examine customer demographics, ride behavior, temporal trends, station performance, trip characteristics, fleet utilization, and relationships between key variables. These analyses revealed valuable insights into rider behavior, demand patterns, and operational performance throughout the year.

The findings indicate that the Ford GoBike system is predominantly used by subscribers for weekday commuting, with demand peaking during morning and evening rush hours. High-traffic transit stations and popular travel routes were identified, while most trips were found to be short in duration, highlighting the system's role in supporting urban mobility and first-mile/last-mile transportation.

Based on these findings, several business recommendations were proposed, including optimizing bike distribution during peak periods, expanding capacity at high-demand stations, improving weekend engagement, implementing predictive maintenance, and leveraging historical data for demand forecasting.

Overall, this project demonstrates how data analytics can transform raw transportation data into actionable business insights. By combining effective data preprocessing, visualization, and analytical reasoning, the study provides meaningful recommendations that can support operational improvements and enhance customer experience. It also showcases practical skills in Python, Pandas, NumPy, Matplotlib, Seaborn, data cleaning, feature engineering, exploratory data analysis, and business storytelling, making it a strong portfolio project for aspiring data analysts.